



NTNU – Trondheim
Norwegian University of
Science and Technology

TET4185: Power Markets, Resources and Environment

Microeconomics

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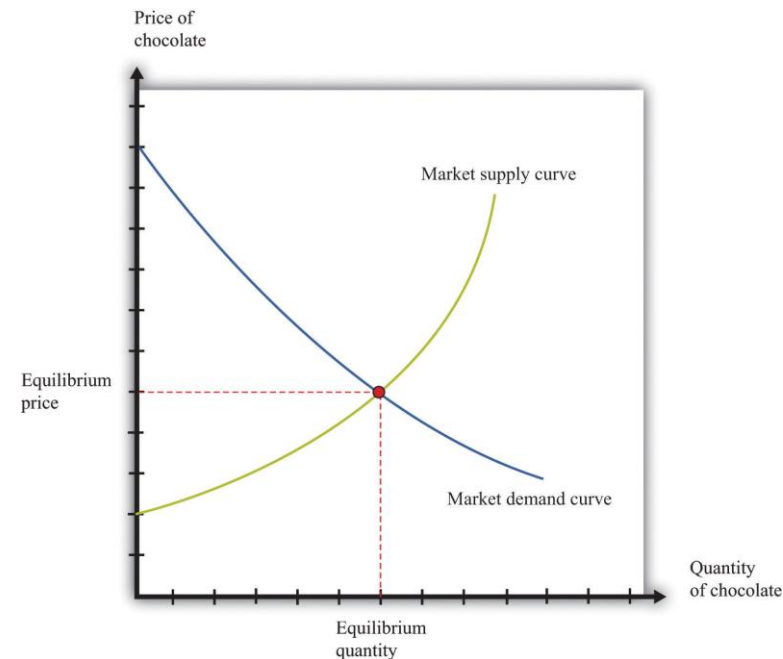


Overview

- Supply, demand, market balance
- Demand price elasticity, direct price elasticity, cross price elasticity
- Social welfare and economic surplus
- Different forms of competition- perfect competition, monopoly, oligopoly
- Competition example-small power system

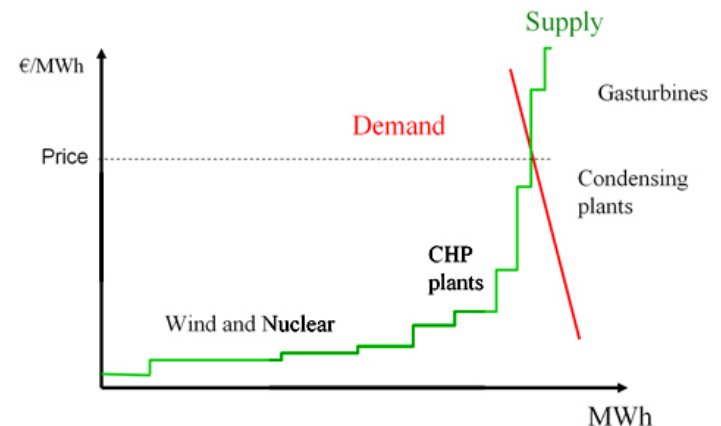
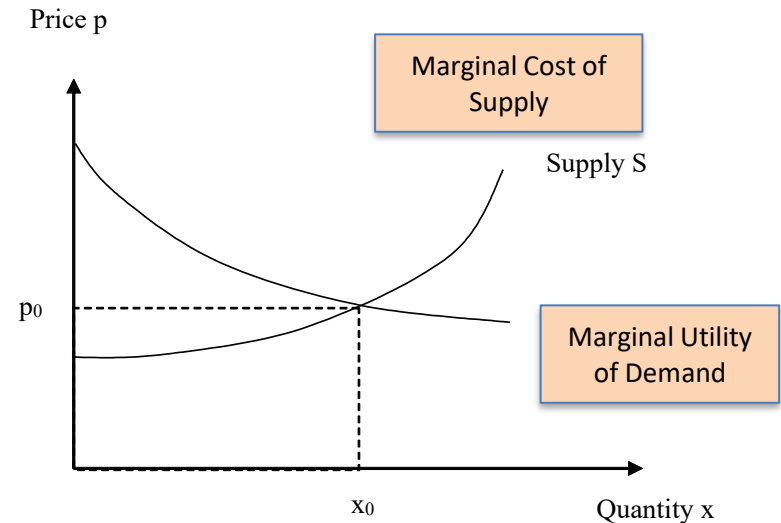
Market prices – how they are «made»

- A market setting considers two essential contributors: Supplier and consumer (Supply and demand)
- Supply of a given resource increases as market price increases
 - Marginal cost of production increases with increasing quantity
 - **Supply curve:** Marginal cost of supplying the resource
- Demand for a resource decreases with higher market price
 - Willingness to buy
 - **Demand curve:** Marginal willingness to buy the resource
- Market balance is where the curves meet



Marshallian supply-demand cross

- Alfred Marshall:
 - «Supply and demand interact to determine market balance, deciding price and quantity»
- The market cross is «equilibrium»
 - One defined price for a resource
- Practically, the trading is done to lead to one unified price



Demand price elasticity



The elasticity of demand shows how demand varies with prices

“How much the quantity demanded of a good changes in response to a change in the price of that good.”

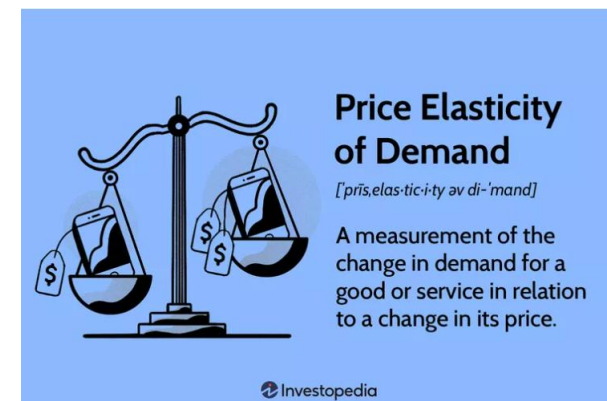
- Elasticity is used to gauge how “Demand depends on price”

- Demand as a function of price: $x = f(p)$

- A small change in x : $\Delta x = \frac{\partial f}{\partial p} \Delta p$

- relative change: $\frac{\Delta x}{x} = \left(\frac{\partial f}{\partial p} \frac{p}{x} \right) \frac{\Delta p}{p} = e \cdot \frac{\Delta p}{p}$ where $e = \frac{\partial f}{\partial p} \frac{p}{x}$

- e is normally negative (Giffen's Paradox, Veblen effect)



Theresa Chiechi / Investopedia

If the elasticity is $e = -0.5$, it means that one per cent increase in price gives 0.5 per cent decrease in quantity.

- Constant elasticity $x = A \cdot p^{-k}$

$$e = \frac{\partial x}{\partial p} \cdot \frac{p}{x} = -\left(k \cdot A \cdot p^{-k-1}\right) \cdot \frac{p}{A \cdot p^{-k}} = -k$$

Direct price elasticity vs Cross-price elasticity

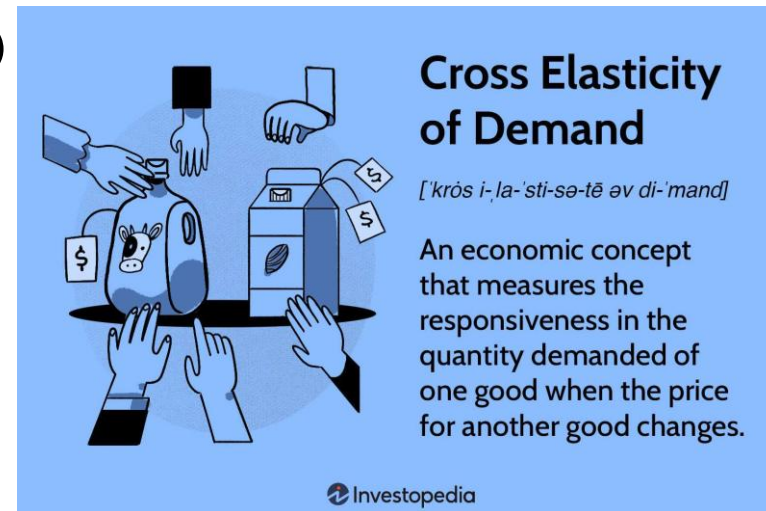
- The demand for a product does not depend on the price of this product alone. It depends on the price of all

products : $x_i = f_i(p_1, p_2, \dots, p_i, \dots, p_n)$

- Direct price elasticity: $e_{11} = \left(\frac{\partial x_1}{\partial p_1} \frac{p_1}{x_1} \right)$

- Cross-price elasticity: $e_{12} = \left(\frac{\partial x_1}{\partial p_2} \frac{p_2}{x_1} \right)$

- Examples: fuel substitution, demand substitution between hours



Inelastic demand

- When demand is fixed, not adjusted by price
- Typically, this can be related to goods that «are necessary»
- What goods are inelastic?
- Electricity was seen in Norway to be elastic during high prices in November 2022 for end-users

Inelastic Demand



«Reduction in power consumption in November 2022 compared to 2021, due to power price difference»

Redusert strømforbruk i november 2022

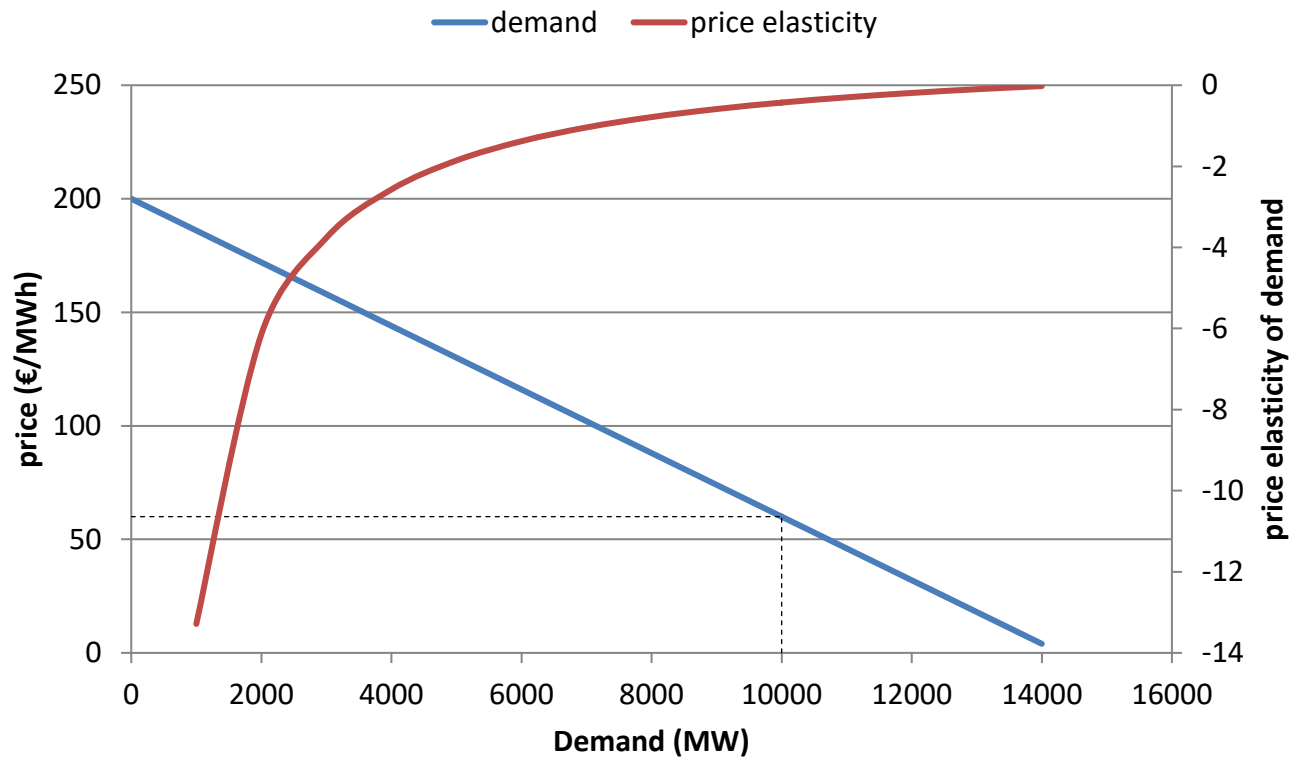
27.12.2022 10:11:00 CET | [Norges vassdrags- og energidirektorat \(NVE\)](#)



Strømforbruket gikk betydelig ned i november sammenlignet med 2021, viser analyser fra NVE. Husholdningene i det sørlige Norge har redusert strømforbruket med 10,7 prosent i november sammenlignet med i fjor. Strømforbruket i fritidsboliger i det sørlige Norge er redusert med hele 23,5 prosent i november sammenlignet med i 2021. Tallene er korrigert for temperaturvariasjoner.

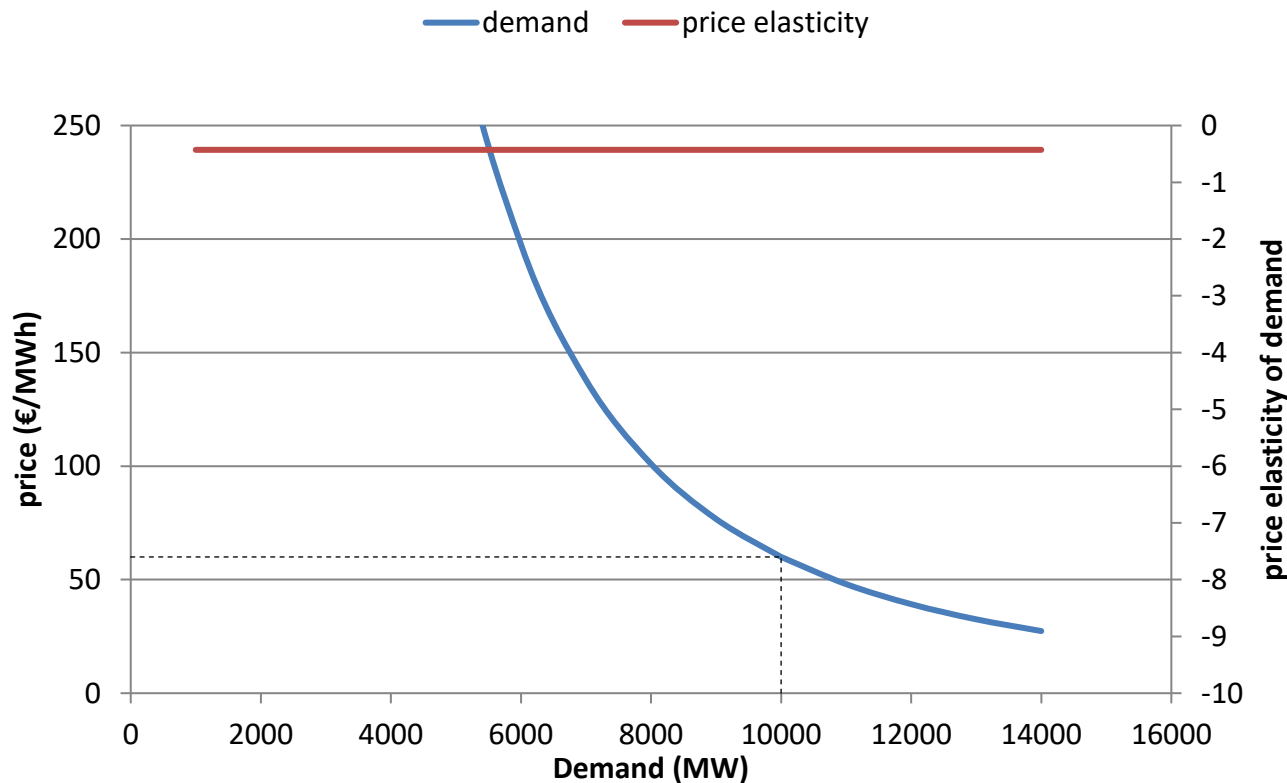
Linear demand and price elasticity

$$p = a - bx \Rightarrow e = 1 - \frac{a}{bx} \quad a = 200, b = 0.014$$

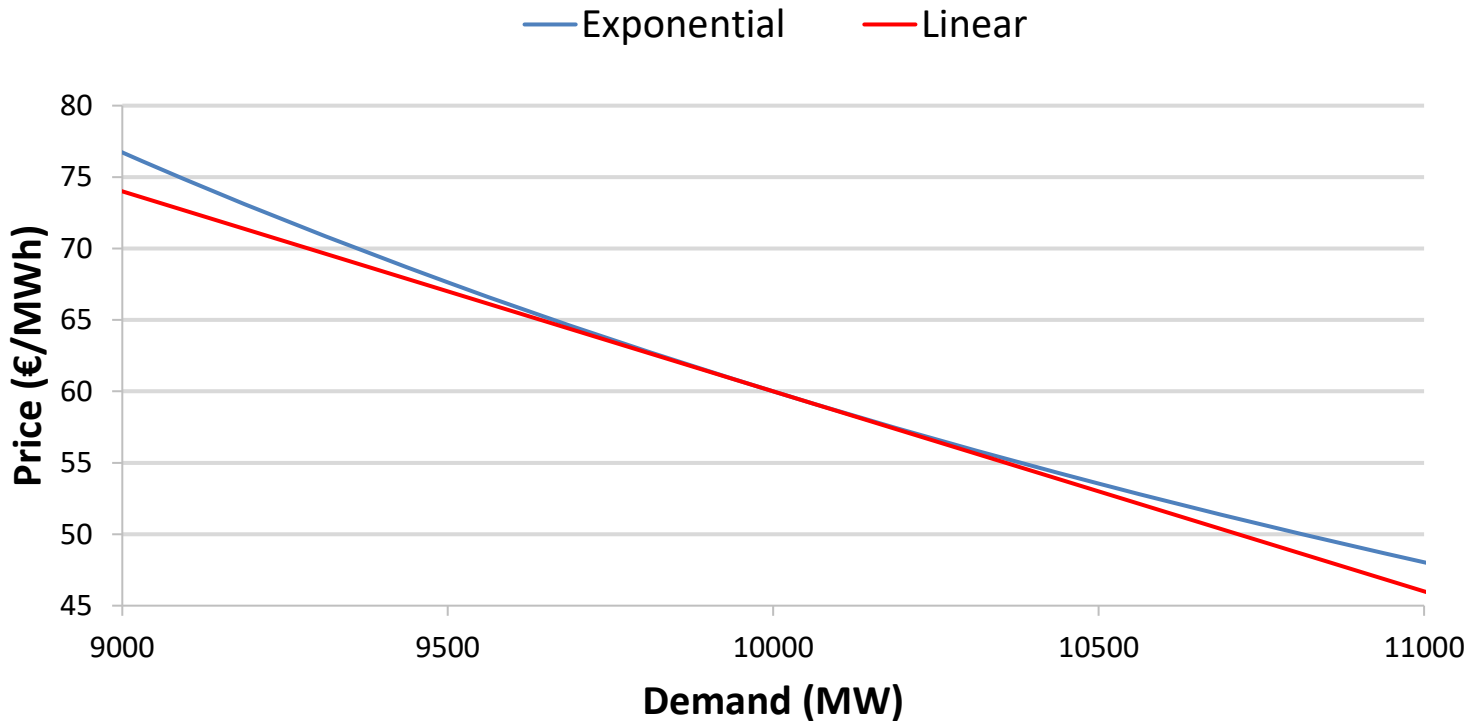


Exponential demand and Constant demand elasticity

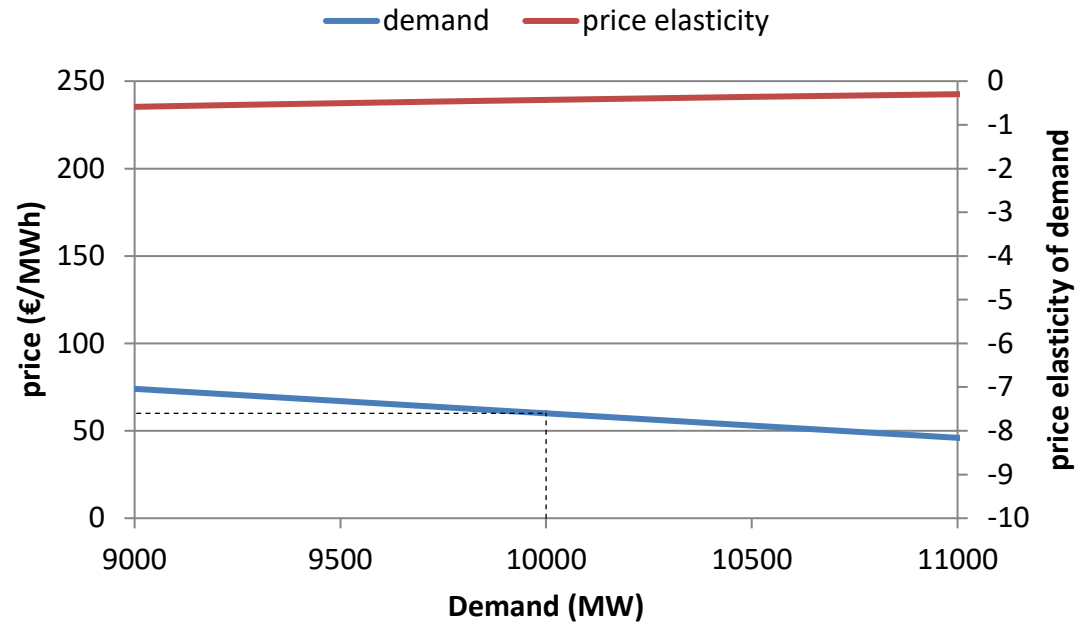
$$x = ap^{-k} \Rightarrow e = -k \quad a = 57818.2, k = 0.42857$$



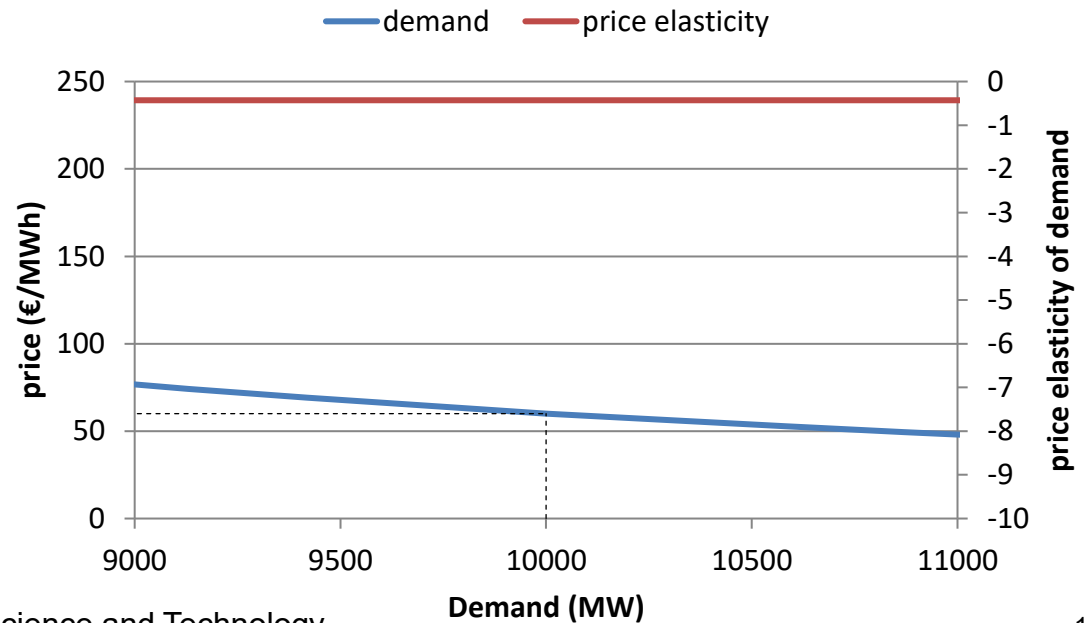
Is the linear model "stupid"?



linear model

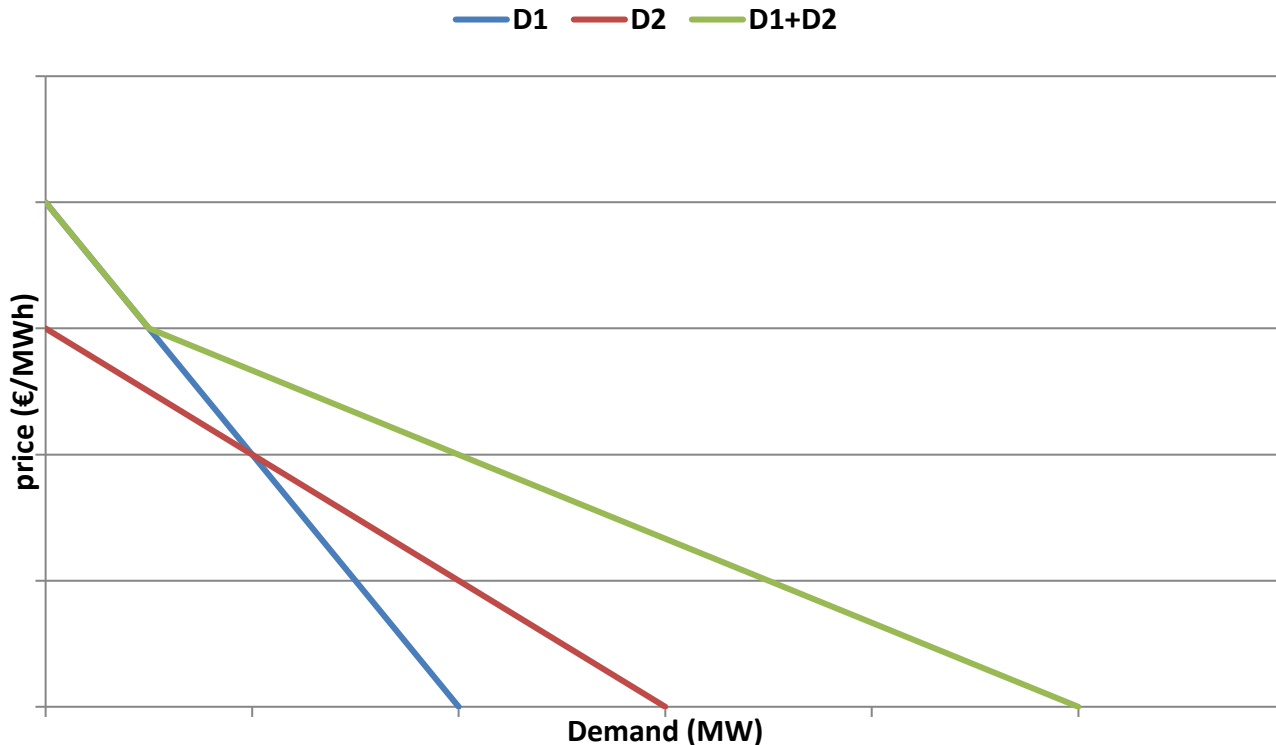


constant elasticity model



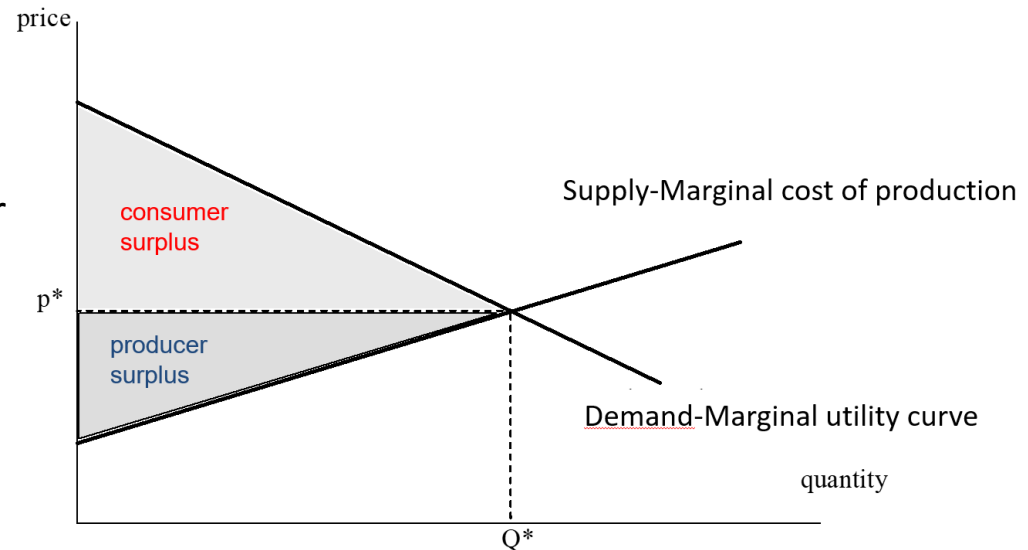
Market Demand Curve-“horizontal sum”

- A market demand curve is the “horizontal sum” of each individual demand curve.
- For each price the total quantity demanded by the market is the sum of the amounts demanded by each individual



Producer and Consumer Surplus – Why we want market equilibrium

- **Producer Surplus (PS):** The benefit gained by selling Q^* for price p^*
 - Accounts for cost of operation
- **Consumer Surplus (CS):** The benefit for the consumers by paying less for the service than their willingness
- **Social welfare (SW):** The total economic surplus to society, the value given to society from the market
- $CS + PS = SW$ (**Social Welfare**)
- Optimal solution for society is to maximize SW, which happens when the two curves cross, defined as «perfect competition market»



Exercise



- Linear demand curve
 - $D = C + f \cdot p$ (MW)
 - $C = 14286, f = -\frac{1}{0.014}$
- Exponential demand curve
 - $D = A \cdot p^{-k}$ (MW)
 - $A = 57818.2, k = 0.42857$
- Calculate
 - Price p (€/MWh) for $D = 9000, 10000, 11000$ (MW)
 - Price elasticity e for same values of D

Solution-Demand Price Elasticity

- Prices, linear model

- $p = 200 - 0.014D$
- $p_{9000} = 74, p_{10000} = 60, p_{11000} = 46$

- Prices, exponential model

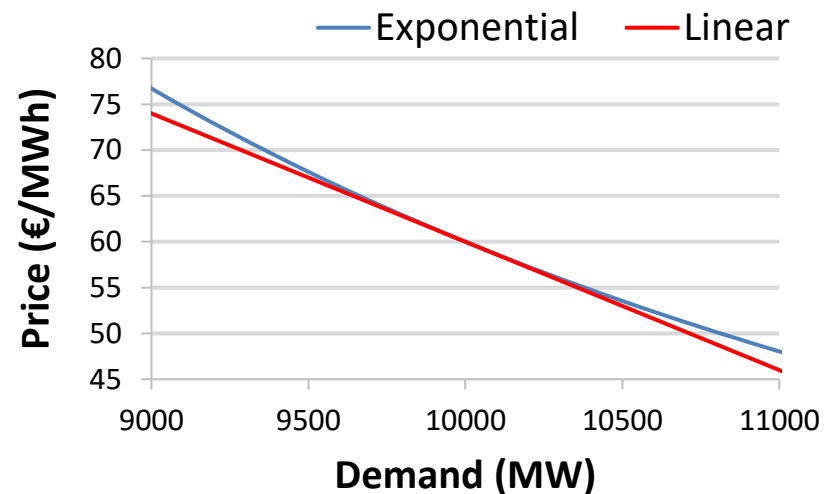
- $p = A^{-1/k} \cdot D^{1/k}$
- $p_{9000} = 76.7, p_{10000} = 60, p_{11000} = 48.0$

- Elasticities, linear model

- $e = \frac{\partial D}{\partial p} \cdot \frac{p}{D} = 1 - \frac{200}{0.014 \cdot D}$
- $e_{9000} = -0.5873, p_{10000} = -0.4286, p_{11000} = -0.2987$

- Elasticities, exponential model

- $e = -0.4286$ (constant)

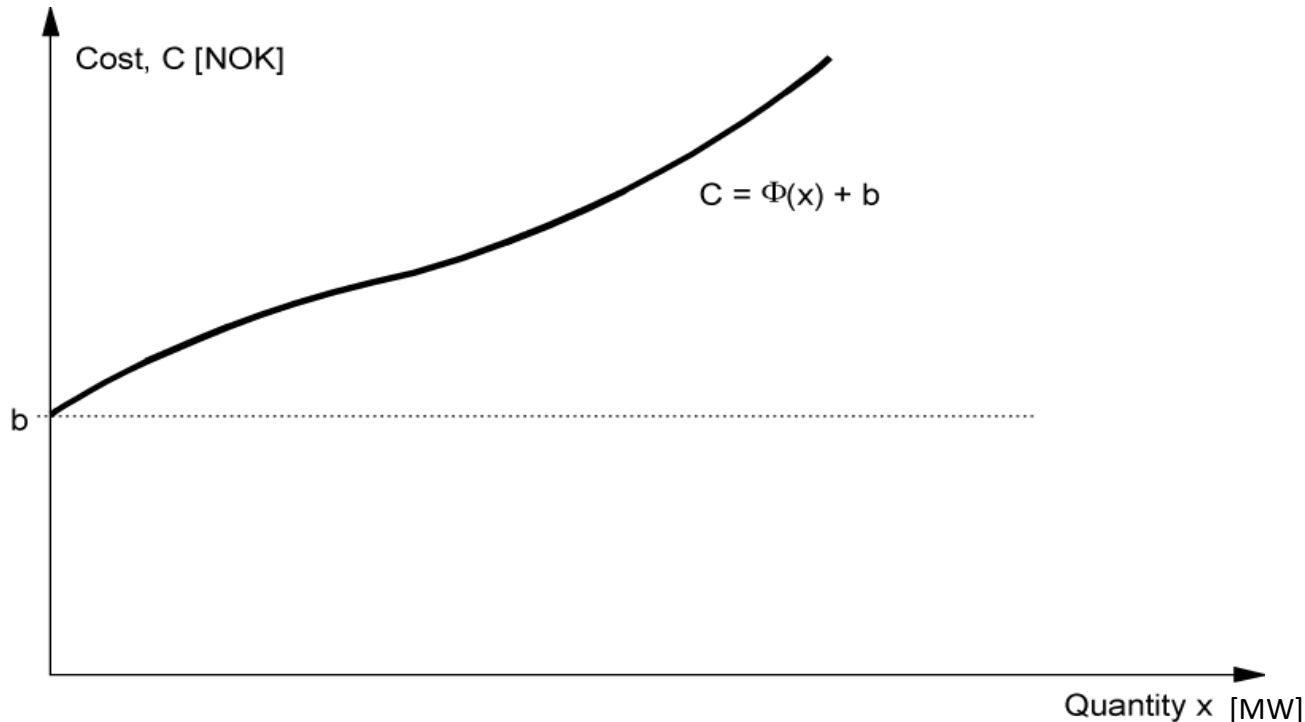


Competition

Perfect competition, Monopoly, Oligopoly



Production Cost



- Production of a resource is dependent on the quantity of the resource produced
- Increasing production increases cost

Production Cost

- Fixed cost: Costs that don't change with the amount produced
- Variable cost: Costs that do change with the amount produced
- Total cost: Fixed cost plus variable cost

The cost of production expressed as a function of the level of output p :

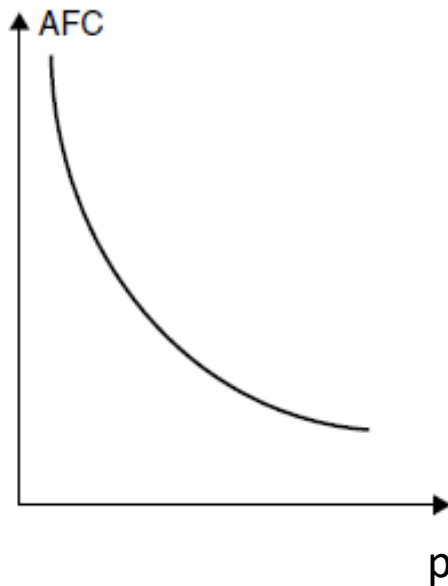
$$c(p) = c_v(p) + c_f$$

- Where $c_v(p)$ represents the variable costs and c_f represents the fixed costs.
- The *average cost function* measures the cost per unit of output

$$AC(p) = \frac{c(p)}{p} = \frac{c_v(p)}{p} + \frac{c_f}{p} = AVC(p) + AFC(p)$$

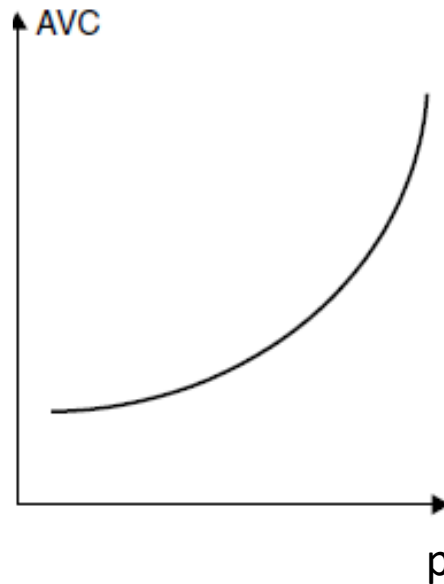
- Where $AVC(p)$ is the *average variable cost* and $AFC(p)$ the *average fixed cost*.

Typical Shapes of the Average Cost Functions



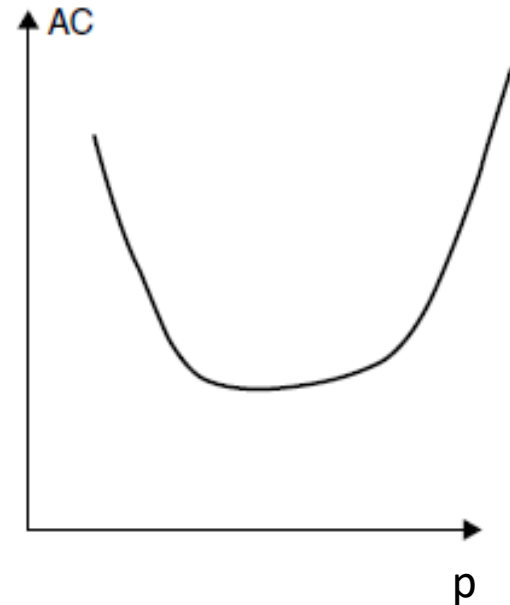
$$AFC(p) = \frac{c_f}{p}$$

+



$$AVC(p) = \frac{c_v(p)}{p}$$

=

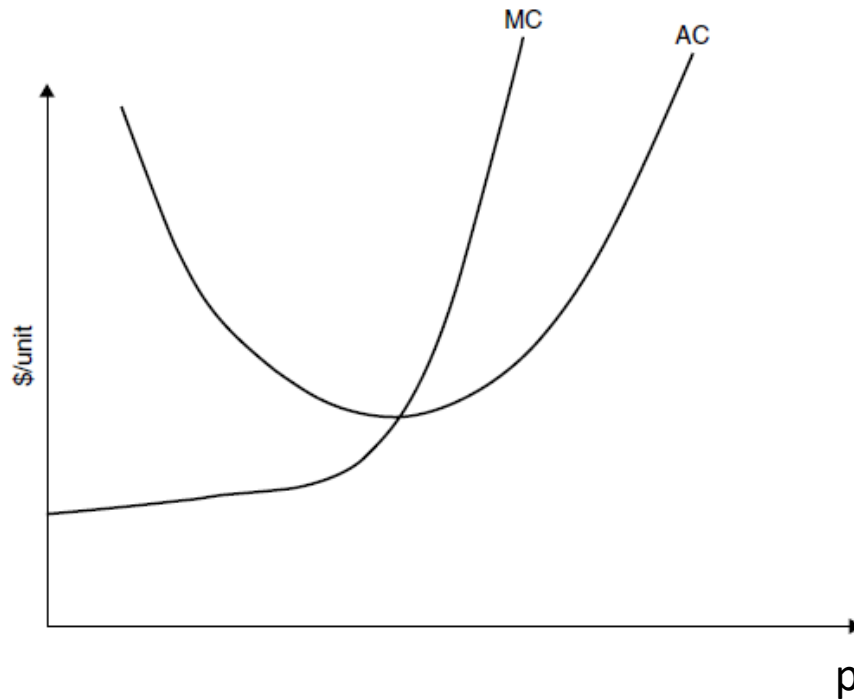


$$AC(p) = AVC(p) + AFC(p)$$

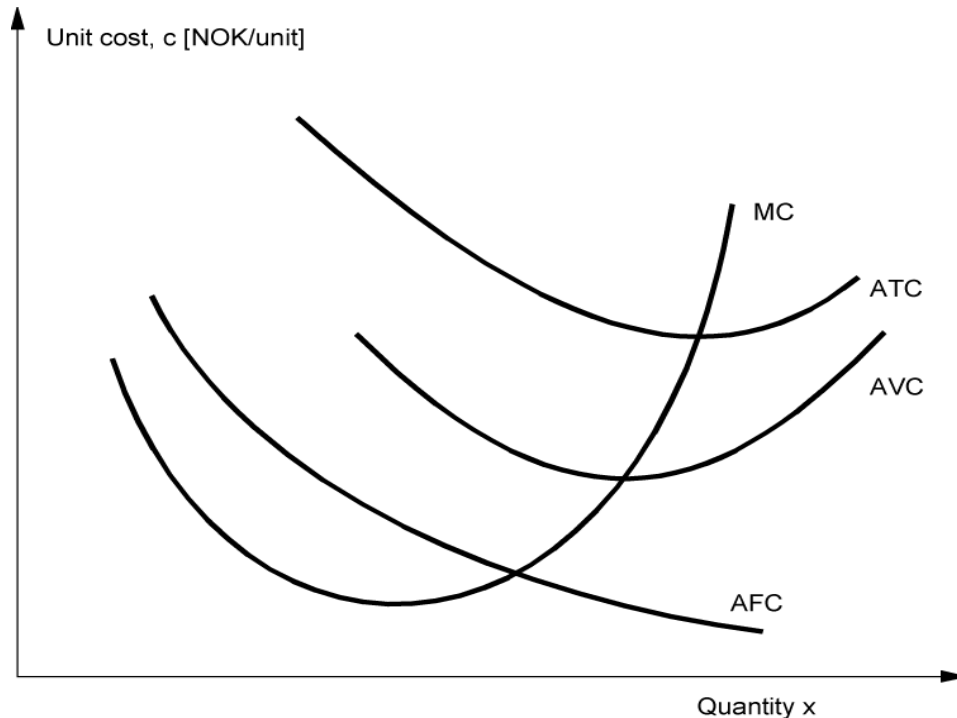
Average Cost vs Marginal cost

- The *marginal cost* reflects the cost of the last unit produced.
- The *average cost* factors are the cost of all the units already produced.

$$MC(p) = \frac{dc(p)}{dp} = \frac{dc_v(p)}{dp}$$



Unit Costs of Production



Average Total Cost:

$$\cancel{ATC = \frac{\Phi(x) + b}{x}}$$

Average Variable Cost:

$$AVC = \frac{\Phi(x)}{x}$$

Average Fixed Cost:

$$\cancel{AFC = \frac{b}{x}}$$

Marginal Cost:

$$MC = \frac{d\Phi(x)}{dx}$$

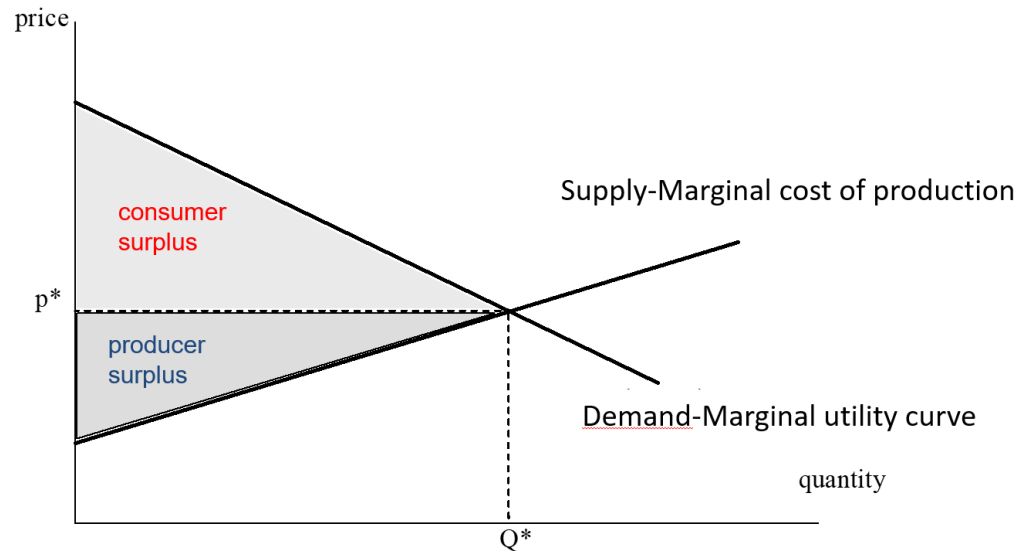
Short-term Decisions

Marginal costs

- Marginal cost: cost for producing one more unit, given the current production level
- Short-term: no new investments, marginal cost based on existing system
- Power production:
 - marginal cost equals the variable cost of the most expensive unit(s) in operation (€/MWh)
 - if demand increases with 1 MW, it is necessary to take this increase in the most expensive unit(s) in operation
 - ... because all other units already are loaded up to their maximum
- Power transport (transmission & distribution)
 - marginal cost equals the marginal cost of delivering power to a specific location in the grid (€/MWh or €/MWyr)
 - ✓ Includes energy delivery, grid losses, outages and O&M costs

How does competition affect the market equilibrium?

- How should producers set the price in the market?
- Producers want to maximize producer surplus
- It depends on the role of the producer:
 - Are you a price taker?
 - Are you a price maker?
 - Are you alone or competing with others?



Perfect competition vs Monopoly

- Perfect competition
(price taker)

cost: $C = \Phi(x) + b$

profit: $\pi = xp_x - (\Phi(x) + b)$

$$\frac{d\pi}{dx} = 0 \text{ and assuming price taker: } p_x = \frac{d\Phi(x)}{dx}$$

$$p_x = \frac{d\Phi(x)}{dx} - x \frac{dp_x}{dx}$$

- Monopoly
(price is a function of quantity)

$$p_x = p_x(x)$$

cost: $C = \Phi(x) + b$

profit: $\pi = xp_x(x) - (\Phi(x) + b)$

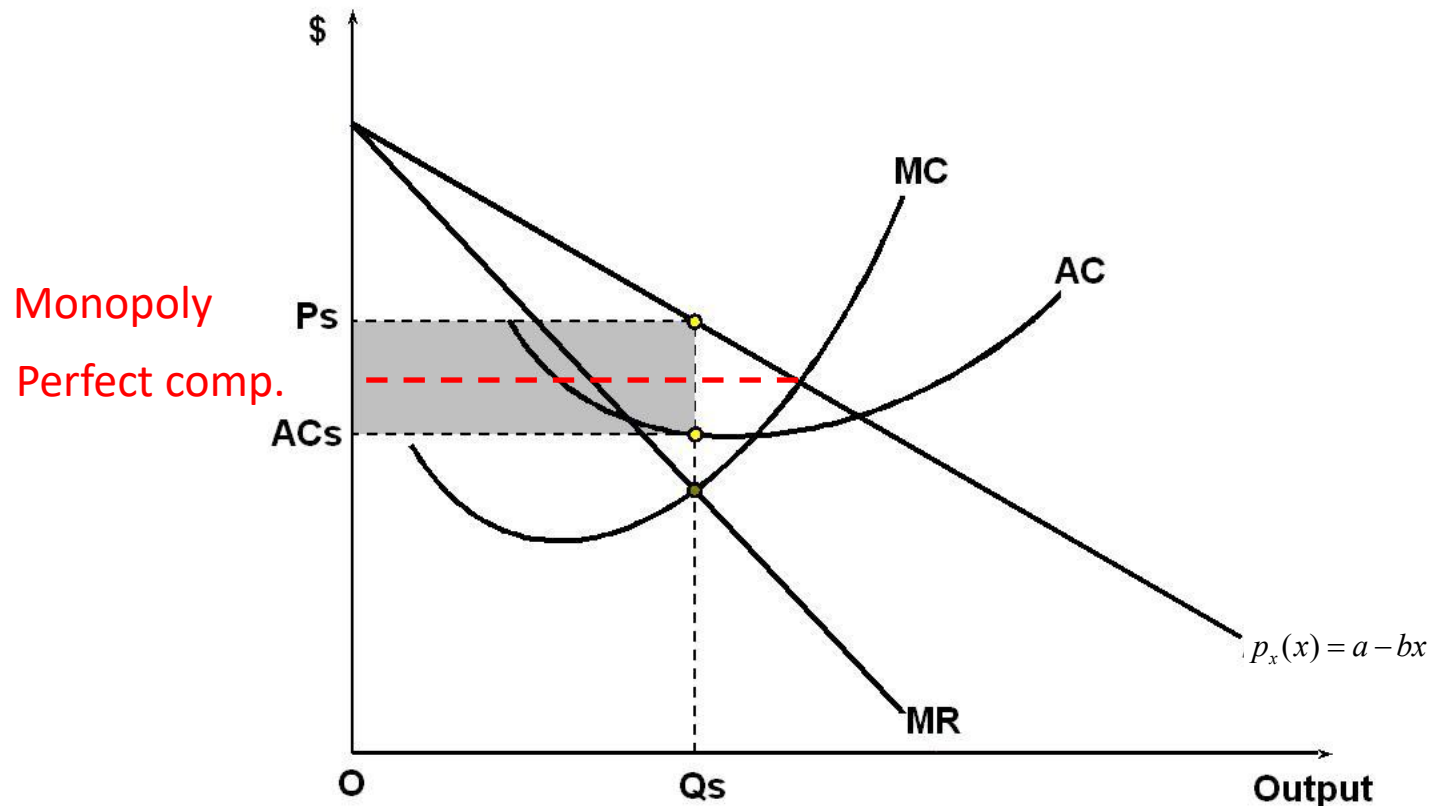
$$\frac{d\pi}{dx} = x \frac{dp_x}{dx} + p_x - \frac{d\Phi(x)}{dx} = 0 \Rightarrow \frac{d\Phi(x)}{dx} = x \frac{dp_x}{dx} + p_x$$

RHS in monopoly is marginal revenue. First term is negative, this means

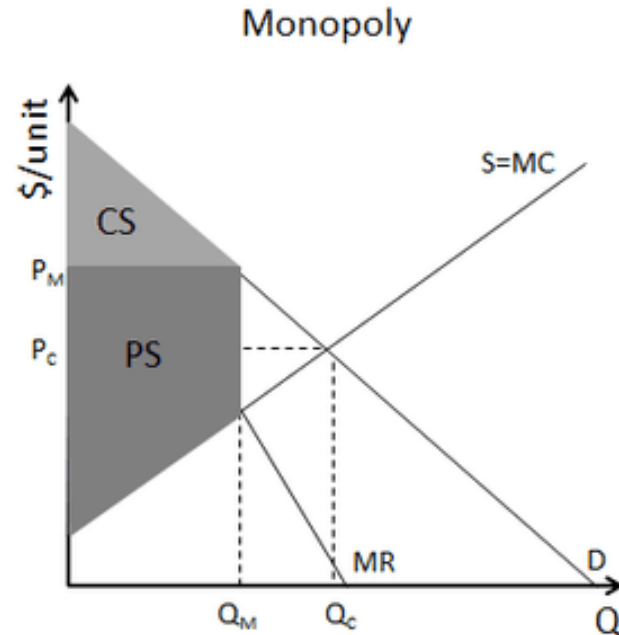
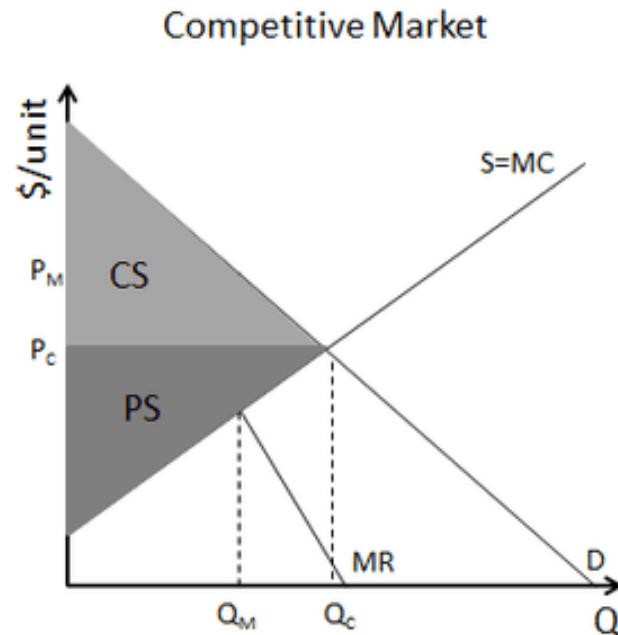
monopoly price > perfect competition price

Monopoly (Linear Price)

$$p_x(x) = a - bx \Rightarrow \text{marginal revenue} = x \frac{dp_x}{dx} + p_x = a - 2bx$$



Perfect competition vs Monopoly – Market result



Source: <https://www.thoughtco.com/the-economic-inefficiency-of-monopoly-1147784>

Social welfare is denoted as $CS + PS$: In which market setting is social welfare maximized? What would society benefit the most from?

Oligopoly, Cournot competition

- Several (few) firms
- Each firm considers the effect of its output decisions on the price
- Assumption: output decisions do not affect other firms' output decisions
- Solution obviously between perfect competition and monopoly
- Variant: some (large) firms compete imperfectly (e.g. Cournot fashion) while others behave as "competitive fringe"
 - And reap benefits of imperfect competition

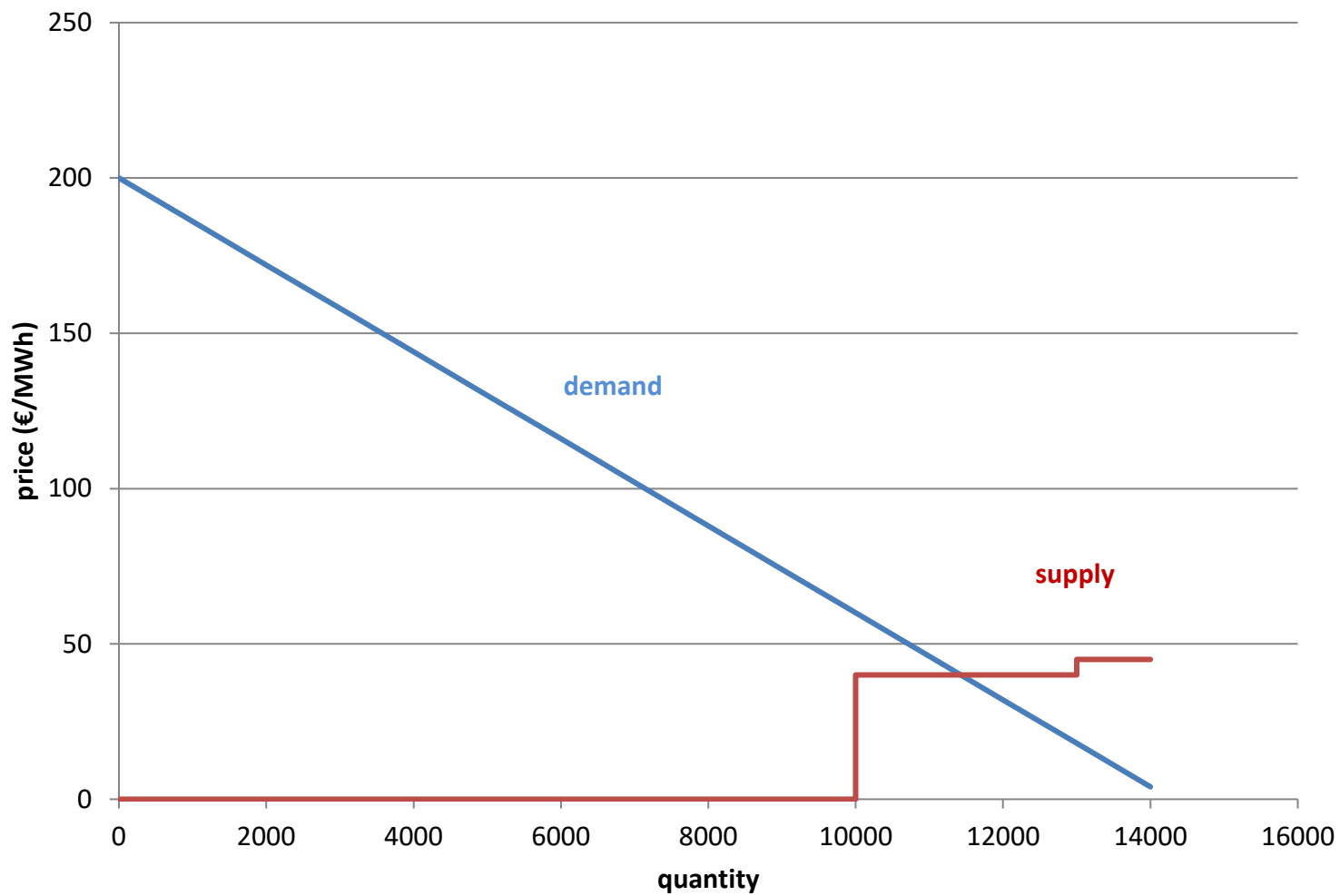
Oligopoly, Cournot competition (Solution)

- For each firm i the profit equals: $\pi_i = p_i \cdot x_i - C_i$ and $C_i = \Phi_i(x_i) + b_i$
- Marginal Revenue of firm i : $\frac{d\pi_i}{dx_i} = p_i + x_i \frac{dp_i}{dx_i} - MC_i(x_i)$
- Assume that the firm's output decisions do not affect those of other firms $\rightarrow$ the price of quantity i is just changed by quantity i : $\frac{dp_i}{dx} = \frac{dp_i}{dx_i}$
- Then you get one equation for each firm, and the sum of quantities equals the demand. By solving this set of equations, you can find the solution

Example



- Small power system
- $P = 200 - 0.014D$
- Three power plants:
 - Hydro, 10000 MW, $C_h = 0$
 - Gas 1, 3000 MW, $C_1 = 40$ €/MWh
 - Gas 2, 3000 MW, $C_2 = 45$ €/MWh
- Find the competitive, monopolistic and Cournot solution



Solution-Competition

- Competitive solution: easy to see from figure, $P = 40 \text{ €/MWh}$

$$D = \frac{200 - 40}{0.014} = 11428.5 \text{ MW}$$

- $\pi = 11428.5 \cdot 40 - 1428.5 \cdot 40 = 400.000 \text{ €}$

- Alternative solution (revenue):

- Revenue: $\pi = P \cdot X - C(X)$

- With $D = X$:
$$\frac{d\pi}{dX} = 0 \Rightarrow 200 - 0.014X = c$$

where

$$c = 0 \text{ for } 0 \leq X \leq 10000$$

$$c = 40 \text{ for } 10000 \leq X \leq 13000$$

$$c = 45 \text{ for } 13000 \leq X \leq 16000$$

Solution-Competition

- Monopoly solution: assume one producer owns everything

$$\pi = P(X) \cdot X - C(X)$$

With $D = X$:

$$\frac{d\pi}{dX} = 0 \Rightarrow 200 - 0.028X = c$$

where

$$c = 0 \text{ for } 0 \leq X \leq 10000$$

$$c = 40 \text{ for } 10000 \leq X \leq 13000$$

$$c = 45 \text{ for } 13000 \leq X \leq 16000$$

- results in $Q = D = 7142.8$ MW and $P = 100$ €/MWh
- $\pi = 7142.8 \cdot 100 = 714280$ €

Solution-Competition

- Cournot solution: assume hydro produces anyway, and two others compete

$$\pi_i = P \cdot q_i - c_i q_i$$

$$\frac{\partial \pi_i}{\partial q_i} = P + q_i \frac{\partial P}{\partial q_i} - c_i = 0$$

this gives:

$$P - 0.014q_2 - 40 = 0$$

$$P - 0.014q_3 - 45 = 0$$

results in:

$$200 - 0.014 \cdot (10000 + q_2 + q_3) - 0.014q_2 - 40 = 0$$

$$200 - 0.014 \cdot (10000 + q_2 + q_3) - 0.014q_3 - 45 = 0$$

this becomes:

$$2q_2 + q_3 = 1428.6$$

$$q_2 + 2q_3 = 1071.4$$

and

$$q_2 = 238.1, q_3 = 595.2, Q = 10833.3 \text{ MW and } P = 48.3 \text{ €/MWh}$$

$$\pi = 48.3 \times 10833.3 - 40 \times 238.1 - 45 \times 595.2 = 486940 \text{ €}$$

